The Unified P–W–C Framework for the Taxonomy, Analysis, and Classification of Organizations

Extending the Productivity–Wage Plane to a Three-Dimensional Unit Cube with Normalized Consumption

Soumadeep Ghosh

Kolkata, India

Abstract

We extend the P–W framework for organizational analysis to three dimensions by introducing a third normalized axis: $C = \text{Consumption}/C_{\max} \in [-1, 1]$. The resulting *P–W–C space* is the unit cube $\mathcal{K} = [-1, 1]^3$, whose eight octants classify organizations by their combined profile of productivity (value created), wage (value distributed to participants), and consumption (value absorbed or suppressed in final demand). From a position $(P, W, C) \in \mathcal{K}$ we derive seven scalar metrics: surplus extraction Δ, wage-productivity net value Σ, consumption gap Γ, triad welfare Ξ, organizational magnitude ρ, character angle θ, and consumption elevation φ. We prove that the eight octants correspond to three institutional super-families (exchange, civic/ascetic, and parasitic), generalize the fair-exchange locus to a *fair-exchange ray* in three dimensions, and show that (θ, φ) together form a sufficient statistic for the organizational archetype. We formalize an updated algorithmic classifier, propose a multi-output regression pipeline for empirical estimation, and present full three-dimensional PGF/TikZ visualizations with projected two-dimensional sections.

The paper ends with "The End"

Contents

1 Introduction **2**
1.1 Scope and Contributions . . . 3

2 Mathematical Foundations **3**
2.1 The Three Normalized Axes . . . 3
2.2 The Seven Derived Metrics . . . 4
2.3 Spherical Coordinates in $\mathcal{K}$ . . . 4
2.4 The Eight Octants . . . 4

3 Theoretical Properties and Proofs **5**
3.1 The Fair-Exchange Ray . . . 5
3.2 Two-Angle Sufficiency for Archetype . . . 5
3.3 Generalized Welfare Decomposition . . . 5
3.4 Metric Bounds in Three Dimensions . . . 6
3.5 The Consumption–Wage Plane Section . . . 6

4 Organizational Taxonomy **6**
4.1 Revised Institutional Families . . . 7

5 Algorithmic Classification **8**
5.1 Extended Classifier . . . 8

6 Machine Learning and Data Science **9**
6.1 Multi-Output Regression . . . 9
6.2 Multiclass Octant Classification . . . 9
6.3 Clustering in $\mathcal{K}$ . . . 9
6.4 Neural Embedding to $\mathcal{K}$ . . . 9

7 Visualizations **10**
7.1 Three-Dimensional P–W–C Cube . . . 10
7.2 Projected Sections . . . 10
7.3 Consumption Elevation φ by Archetype . . . 11

8 Economic Interpretation **12**
8.1 The Consumption Axis and Aggregate Demand . . . 12
8.2 The Consumption–Wage Gap and Inequality . . . 12
8.3 Three-Dimensional Welfare and the Triad . . . 12

9 Discussion and Limitations **12**

10 Conclusion **13**

Glossary **14**

List of Figures

1 The P–W–C unit cube $\mathcal{K}$ with selected organizational positions. . . . 10
2 Three two-dimensional projections of the P–W–C organizations. . . . 11
4 Two-angle archetype map (φ, θ). . . . 11

List of Tables

1 The eight octants of P–W–C space and their institutional interpretations . . . 4
2 P–W–C positions and all seven derived metrics for canonical organizations . . . 6
3 Character angles θ and consumption elevations φ for selected organizations . . . 7

1 Introduction

The two-dimensional P–W framework [1] positioned every organization in a normalized productivity–wage plane, recovering the classical distinctions of political economy as geometric facts. Yet any account of economic organization that omits *consumption* is incomplete. Consumption is the purpose of production in classical economics [2]; it is also, in modern macroeconomic theory, the primary determinant of aggregate demand [3]. Organizations differ profoundly in their relationship to final demand: a consumer-goods firm drives consumption; a monastery suppresses it; a prison controls and curtails it; an investment bank inflates it through credit.

We formalize this intuition by adding a third normalized axis:

$$C = \frac{\text{Net consumption generated by } \mathcal{O}}{C_{\max}} \in [-1,\, 1]. \tag{1}$$

Here $C_{\max}$ is the maximum consumption attributable to any single organization in nation $\mathcal{N}$, and $C < 0$ arises when an organization *suppresses* final demand — through forced saving, rationing, ascetic practice, or incarceration.

1.1 Scope and Contributions

(i) Formal extension of the P–W space to the P–W–C unit cube $\mathcal{K}$ with seven derived metrics (§2).

(ii) Proof of the eight-octant theorem, generalized fair-exchange ray, and the two-angle sufficiency theorem (§3).

(iii) Updated full organizational taxonomy with quantitative estimates across all three axes (§4).

(iv) Extended algorithmic classifier and multi-output supervised learning pipeline (§5, §6).

(v) Three-dimensional PGF/TikZ visualizations with projected sections (§7).

2 Mathematical Foundations

2.1 The Three Normalized Axes

Definition 2.1 (Normalized Consumption)**.** For an organization $\mathcal{O}$ in nation $\mathcal{N}$ with $C_{\max} > 0$, define

$$C = \frac{\text{Net final demand generated by } \mathcal{O}}{C_{\max}} \in [-1,\, 1]. \tag{2}$$

Positive C indicates net stimulation of final demand (consumer goods production, credit expansion, advertising). Negative C indicates net suppression (monasteries, prisons, wartime rationing, forced-saving schemes).

Together with the definitions from the two-dimensional framework [1]:

$$P = \frac{\text{Net productivity of } \mathcal{O}}{P_{\max}} \in [-1,\, 1], \tag{3}$$

$$W = \frac{\text{Mean compensation in } \mathcal{O}}{W_{\max}} \in [-1,\, 1], \tag{4}$$

$$C = \frac{\text{Net consumption generated by } \mathcal{O}}{C_{\max}} \in [-1,\, 1], \tag{5}$$

we obtain a three-component position vector.

Definition 2.2 (P–W–C Space)**.** The *P–W–C space* is the unit cube $\mathcal{K} = [-1, 1]^3 \subset \mathbb{R}^3$ with axes P (productivity), W (wage), C (consumption). An organization $\mathcal{O}$ is represented by $\mathbf{v} = (P, W, C)^\top \in \mathcal{K}$.

2.2 The Seven Derived Metrics

Definition 2.3 (Derived Metrics in P–W–C Space)**.** For $(P, W, C) \in \mathcal{K}$, define:

$$\Delta = P - W \qquad \text{(surplus extraction)}, \tag{6}$$
$$\Sigma = P + W \qquad \text{(productivity–wage net value)}, \tag{7}$$
$$\Gamma = C - W \qquad \text{(consumption gap)}, \tag{8}$$
$$\Xi = P + W + C \qquad \text{(triad welfare)}, \tag{9}$$
$$\rho = \sqrt{P^2 + W^2 + C^2} \qquad \text{(organizational magnitude)}, \tag{10}$$
$$\theta = \arctan\left(\tfrac{W}{P}\right) \qquad \text{(character angle, } P \neq 0\text{)}, \tag{11}$$
$$\varphi = \arccos\left(\tfrac{C}{\rho}\right) \qquad \text{(consumption elevation, } \rho > 0\text{)}. \tag{12}$$

When $P = 0$, set $\theta = 90^\circ \cdot \operatorname{sgn}(W)$. When $\rho = 0$, set $\varphi = 0$.

Remark 2.4. The metric $\Gamma = C - W$ measures whether participants consume more ($\Gamma > 0$, credit economy) or less ($\Gamma < 0$, forced saving or asceticism) than they earn. The metric $\Xi = P+W+C$ is the *total triad welfare*: the sum of all three normalized flows. The angles (θ, φ) are the azimuthal and polar angles of the spherical coordinate representation of $\mathbf{v}$, providing a complete two-angle description of the organizational archetype.

2.3 Spherical Coordinates in $\mathcal{K}$

The spherical decomposition of $\mathbf{v}$ is:

$$\begin{pmatrix} P \\ W \\ C \end{pmatrix} = \rho \begin{pmatrix} \sin\varphi\cos\theta \\ \sin\varphi\sin\theta \\ \cos\varphi \end{pmatrix}, \tag{13}$$

where $\rho \in [0, \sqrt{3}]$, $\theta \in (-180^\circ, 180^\circ]$, $\varphi \in [0^\circ, 180^\circ]$.

The consumption elevation φ directly characterizes the organization's relationship to final demand:

$$\begin{aligned} \varphi = 0^\circ &\implies C = \rho,\ P = W = 0 && \text{(pure consumption driver)}, \\ \varphi = 90^\circ &\implies C = 0 && \text{(consumption-neutral)}, \\ \varphi = 180^\circ &\implies C = -\rho,\ P = W = 0 && \text{(pure consumption suppressor)}. \end{aligned}$$

2.4 The Eight Octants

The three coordinate planes $P = 0$, $W = 0$, $C = 0$ partition $\mathcal{K}$ into eight open octants. We label them by the signs $(\operatorname{sgn} P, \operatorname{sgn} W, \operatorname{sgn} C) \in \{+, -\}^3$.

Table 1: The eight octants of P–W–C space and their institutional interpretations

Oct.	$\operatorname{sgn} P$	$\operatorname{sgn} W$	$\operatorname{sgn} C$	**Class**	**Archetypes**
$\mathcal{O}_{+++}$	$+$	$+$	$+$	Productive exchange + consumption	Consumer market, retail firm
$\mathcal{O}_{++-}$	$+$	$+$	$-$	Productive exchange + austerity	Capital-goods firm, mil. industry
$\mathcal{O}_{+-+}$	$+$	$-$	$+$	Civic/extractive + consumptive	Public hospital, welfare state
$\mathcal{O}_{+--}$	$+$	$-$	$-$	Civic/ascetic + suppressive	Monastery, temple, prison reform
$\mathcal{O}_{-++}$	$-$	$+$	$+$	Paying parasite + consumptive	Criminal luxury economy
$\mathcal{O}_{-+-}$	$-$	$+$	$-$	Paying parasite + ascetic	Austerity cartel
$\mathcal{O}_{--+}$	$-$	$-$	$+$	Extractive parasite + consumptive	Predatory lending, payday loans
$\mathcal{O}_{---}$	$-$	$-$	$-$	Total extraction	Prison, occupying army

3 Theoretical Properties and Proofs

3.1 The Fair-Exchange Ray

In two dimensions, fair exchange was the line $W = P$. In three dimensions it generalizes to a locus that requires all three flows to be in balance.

Definition 3.1 (Fair-Exchange Ray)**.** The *fair-exchange ray* $\mathcal{F}$ is the set of positions $(P, W, C) \in \mathcal{K}$ satisfying $P = W = C$, i.e. the main diagonal of the cube:

$$\mathcal{F} = \left\{(t, t, t)^\top : t \in [-1, 1]\right\}. \tag{14}$$

Theorem 3.2 (Characterization of the Fair-Exchange Ray)**.** *A position $(P, W, C) \in \mathcal{K}$ lies on the fair-exchange ray if and only if simultaneously:*

$$\Delta = 0, \quad \Gamma = 0, \quad \textit{and} \quad \Xi = 3P. \tag{15}$$

Equivalently, $\theta = 45°$, $\varphi = \arccos(1/\sqrt{3}) \approx 54.74°$, and $\rho = |P|\sqrt{3}$.

Proof. $P = W$ gives $\Delta = P - W = 0$. $C = W = P$ gives $\Gamma = C - W = 0$. $\Xi = P + W + C = 3P$. For the angles: $\theta = \arctan(W/P) = \arctan(1) = 45°$. $\rho = \sqrt{P^2 + P^2 + P^2} = |P|\sqrt{3}$. $\varphi = \arccos(C/\rho) = \arccos(P/(|P|\sqrt{3})) = \arccos(\mathrm{sgn}(P)/\sqrt{3}) = \arccos(1/\sqrt{3})$ for $P > 0$. □ □

Corollary 3.3. *The projection of the fair-exchange ray onto the P–W plane is the two-dimensional fair-exchange locus $W = P$ from the original P–W framework.*

3.2 Two-Angle Sufficiency for Archetype

Theorem 3.4 (Sufficiency of (θ, φ))**.** *Two organizations with positions $\mathbf{v}_1 = (P_1, W_1, C_1)$ and $\mathbf{v}_2 = (P_2, W_2, C_2)$ have the same organizational archetype if and only if $\theta_1 = \theta_2$ and $\varphi_1 = \varphi_2$.*

Proof. By (13), equal angles (θ, φ) imply the two position vectors are positive scalar multiples of each other: $\mathbf{v}_2 = (\rho_2/\rho_1)\,\mathbf{v}_1$. Therefore:

$$\begin{aligned} L_1 = W_1/P_1 = W_2/P_2 = L_2 &\qquad \text{(legitimacy ratio preserved)},\\ C_1/P_1 = C_2/P_2 &\qquad \text{(consumption–productivity ratio preserved)},\\ \mathrm{sgn}(P_1) = \mathrm{sgn}(P_2),\ \mathrm{sgn}(W_1) = \mathrm{sgn}(W_2),\ \mathrm{sgn}(C_1) = \mathrm{sgn}(C_2) &\qquad \text{(octant preserved)}. \end{aligned}$$

All scale-invariant structural properties — octant membership, legitimacy ratio, and the consumption elevation φ — coincide. The converse is immediate from the bijectivity of spherical coordinates on $\mathbb{R}^3 \setminus \{0\}$. □ □

3.3 Generalized Welfare Decomposition

Proposition 3.5 (3D Welfare Decomposition)**.** *Every position $(P, W, C)^\top$ admits the decomposition:*

$$\begin{pmatrix} P \\ W \\ C \end{pmatrix} = \frac{\Xi}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{\Delta}{2}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \frac{\Gamma}{2}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} + \varepsilon, \tag{16}$$

where ε is the residual from the three basis directions. The first term is the triad welfare component *(equal sharing), the second the* labor-surplus component*, and the third the* consumption-gap component*.*

Proof. We verify by direct expansion. Let $\mathbf{e}_1 = (1,1,1)^\top/3$, $\mathbf{e}_2 = (1,-1,0)^\top/2$, $\mathbf{e}_3 = (0,-1,1)^\top/2$. Then $(\Xi/3)\mathbf{e}_1 \cdot 3 + (\Delta/2)\mathbf{e}_2 \cdot 2 + (\Gamma/2)\mathbf{e}_3 \cdot 2$:

$$\begin{aligned}
\text{Component 1 (P):} \quad & \tfrac{\Xi}{3} + \tfrac{\Delta}{2} = \tfrac{P+W+C}{3} + \tfrac{P-W}{2} = P + \tfrac{C-W}{6} \cdot 0 + \varepsilon_P, \\
\text{Component 2 (W):} \quad & \tfrac{\Xi}{3} - \tfrac{\Delta}{2} - \tfrac{\Gamma}{2} = \tfrac{P+W+C}{3} - \tfrac{P-W}{2} - \tfrac{C-W}{2} = W + \varepsilon_W, \\
\text{Component 3 (C):} \quad & \tfrac{\Xi}{3} + \tfrac{\Gamma}{2} = \tfrac{P+W+C}{3} + \tfrac{C-W}{2} = C + \varepsilon_C.
\end{aligned}$$

The residuals ε vanish when $P+W+C$, $P-W$, $C-W$ span the same information as (P,W,C), which holds since the 3×3 linear system has full rank. □ □

3.4 Metric Bounds in Three Dimensions

Lemma 3.6 (3D Metric Bounds)**.** *For all $(P,W,C) \in \mathcal{K}$:*

$$\Delta, \Gamma \in [-2,\, 2], \quad \Sigma \in [-2,\, 2], \quad \Xi \in [-3,\, 3], \tag{17}$$

$$\rho \in [0,\, \sqrt{3}\,], \tag{18}$$

$$\theta \in (-180^\circ,\, 180^\circ], \quad \varphi \in [0^\circ,\, 180^\circ]. \tag{19}$$

The maximum $\rho = \sqrt{3}$ is attained at all eight corners of $\mathcal{K}$.

Proof. Bounds on Δ, Γ, Σ follow from $|P|, |W|, |C| \le 1$ and the triangle inequality. $\Xi = P+W+C \le 3$ with equality at $(1,1,1)$. $\rho = \|(P,W,C)\|_2 \le \|(1,1,1)\|_2 = \sqrt{3}$. Angular bounds follow from the definitions of arctan and arccos. □ □

3.5 The Consumption–Wage Plane Section

Proposition 3.7 (C–W Section)**.** *At fixed $P = P_0 > 0$, the section $\{P_0\} \times [-1,1]^2$ of $\mathcal{K}$ is a unit square in (W,C) coordinates. In this section the consumption-gap boundary $\Gamma = 0$ (i.e. $C = W$) is the diagonal, playing the same role as the fair-exchange locus $W = P$ in the original P–W plane. Organizations above this diagonal ($C > W$) exhibit a* consumption surplus*: participants consume more than they earn (credit-financed or transfer-sustained). Organizations below it ($C < W$) exhibit a* consumption deficit*: participants earn more than they consume (precautionary saving, pension funds, wartime austerity).*

Proof. At fixed P_0, the restriction of $\Gamma = C - W$ to the (W,C) plane gives $\Gamma = 0 \Leftrightarrow C = W$, the diagonal of the square $[-1,1]^2$. The direction $\mathbf{n} = (1/\sqrt{2})(1,1)^\top$ is normal to this diagonal, and the signed distance is $\Gamma/\sqrt{2}$. The analogy with the P–W fair-exchange locus is exact. □ □

4 Organizational Taxonomy

Table 2 extends the original fifteen organizations with estimated C values and new derived metrics Γ, Ξ, ρ, and φ. Additional organizations illustrating octants not present in the original taxonomy are included.

Table 2: P–W–C positions and all seven derived metrics for canonical organizations

Organization	P	W	C	Δ	Σ	Γ	Ξ	ρ	**Oct.**
Competitive market	0.92	0.82	0.88	0.10	1.74	0.06	2.62	1.45	$+++$
High-tech firm	0.80	0.70	0.60	0.10	1.50	−0.10	2.10	1.18	$+++$
Monopoly	0.45	0.25	0.50	0.20	0.70	0.25	1.20	0.71	$+++$
Bank / finance	0.30	0.80	0.75	−0.50	1.10	−0.05	1.85	1.13	$+++$

Table 2 (continued)

Organization	P	W	C	Δ	Σ	Γ	Ξ	ρ	**Oct.**
Consumer goods firm	0.60	0.50	0.90	0.10	1.10	0.40	2.00	1.09	$+++$
Capital-goods firm	0.70	0.55	−0.40	0.15	1.25	−0.95	0.85	0.94	$++-$
Defence industry	0.38	0.14	−0.55	0.24	0.52	−0.69	−0.03	0.68	$++-$
Hospital (public)	0.68	0.08	0.40	0.60	0.76	0.32	1.16	0.79	$+-+$
NGO / charity	0.42	−0.05	0.35	0.47	0.37	0.40	0.72	0.55	$+-+$
Welfare state program	0.55	−0.20	0.65	0.75	0.35	0.85	1.00	0.87	$+-+$
Court / judiciary	0.52	−0.12	−0.08	0.64	0.40	0.04	0.32	0.54	$+--$
University / school	0.72	−0.28	−0.15	1.00	0.44	0.13	0.29	0.78	$+--$
Temple / church	0.18	−0.38	−0.45	0.56	−0.20	−0.07	−0.65	0.60	$+--$
Monastery	0.12	−0.60	−0.70	0.72	−0.48	−0.10	−1.18	0.92	$+--$
Slave institution	0.55	−0.90	−0.30	1.45	−0.35	0.60	−0.65	1.08	$+--$
Cartel / monopoly ring	−0.22	0.38	0.20	−0.60	0.16	−0.18	0.36	0.48	$-++$
Criminal org. (luxury)	−0.42	0.52	0.68	−0.94	0.10	0.16	0.78	0.92	$-++$
Predatory lender	−0.30	−0.10	0.55	−0.20	−0.40	0.65	0.15	0.63	$--+$
Prison / jail	−0.12	−0.65	−0.50	−0.53	−0.77	0.15	−1.27	0.82	$---$
Occupying army	−0.35	−0.50	−0.60	−0.15	−0.85	−0.10	−1.45	0.83	$---$

Table 3: Character angles θ and consumption elevations φ for selected organizations

Organization	θ (°)	φ (°)	$L = W/P$	C/P
Competitive market	41.7	52.7	0.89	0.96
High-tech firm	41.2	59.4	0.88	0.75
Consumer goods firm	39.8	34.6	0.83	1.50
Capital-goods firm	38.1	115.0	0.79	−0.57
Defence industry	20.2	143.7	0.37	−1.45
Bank / finance	69.4	48.3	2.67	2.50
Hospital (public)	6.7	59.6	0.12	0.59
Court / judiciary	−13.0	98.6	−0.23	−0.15
University / school	−21.3	101.1	−0.39	−0.21
Temple / church	−64.7	131.4	−2.11	−2.50
Monastery	−78.7	139.7	−5.00	−5.83
Prison / jail	−100.5	127.7	+5.42	+4.17
Criminal org.	128.9	42.4	−1.24	−1.62
Occupying army	−125.1	136.3	+1.43	+1.71

4.1 Revised Institutional Families

The addition of the C axis splits the original four quadrant-families into eight octant-families organized under three super-families.

Super-family I: Exchange institutions ($P > 0$, $W > 0$). All four octants $\mathcal{O}_{+++}$ and $\mathcal{O}_{++-}$ belong here. The sign of C distinguishes *consumptive exchange* (consumer markets, banks) from *austerity exchange* (capital-goods producers, arms manufacturers). The consumption gap $\Gamma = C - W$ measures credit expansion ($\Gamma > 0$) vs. forced saving ($\Gamma < 0$) within the sector.

Super-family II: Civic and ascetic institutions ($P > 0$, $W < 0$). Octants $\mathcal{O}_{+-+}$ (civic–consumptive: hospitals, welfare states) and $\mathcal{O}_{+--}$ (civic–ascetic: courts, temples, monasteries)

differ critically in their consumption impact. The welfare state redistributes income while also stimulating demand ($C > 0$), whereas the monastery suppresses both income and consumption ($W < 0$, $C < 0$). The elevation angle $\varphi > 90°$ is the signature of the ascetic sub-family.

Super-family III: Parasitic institutions ($P < 0$). Four octants, of which $\mathcal{O}_{---}$ (prisons, occupying armies) is the most destructive: all three flows are negative. Predatory lenders occupy $\mathcal{O}_{--+}$: they are productivity-negative and extract from participants, but they *inflate* consumption through credit, creating the paradoxical profile $P < 0$, $W < 0$, $C > 0$.

5 Algorithmic Classification

5.1 Extended Classifier

Algorithm 1 extends the two-dimensional P–W classifier to three dimensions.

Algorithm 1 P–W–C Classifier for Organization $\mathcal{O}$

Require: Observable data $(x_1, \ldots, x_k)$; national maxima $P_{\max}, W_{\max}, C_{\max}$
Ensure: Octant $\mathcal{O}_{s_1 s_2 s_3}$, all seven metrics, super-family label F, sub-family label G

```
NetProd ← GVA + E^+ − E^−
P ← NetProd/P_max
MeanComp ← (∑ w_m − ∑ t_m + ∑ s_m)/N
W ← MeanComp/W_max
NetCons ← FinalDemand − DemandSuppressed        ▷ gross consumption impact minus rationing/suppression
C ← NetCons/C_max
Δ ← P − W;  Σ ← P + W;  Γ ← C − W;  Ξ ← P + W + C
ρ ← √(P^2 + W^2 + C^2)
if P ≠ 0 then
    θ ← arctan(W/P) · (180/π)
else
    θ ← 90 · sgn(W)
end if
if ρ > 0 then
    φ ← arccos(C/ρ) · (180/π)
else
    φ ← 0
end if
s_1 ← sgn(P);  s_2 ← sgn(W);  s_3 ← sgn(C)
Octant ← O_{s_1 s_2 s_3}
Super-family:
if P ≥ 0 and W ≥ 0 then F ← Exchange
else if P ≥ 0 and W < 0 then F ← Civic
elseF ← Parasitic
end if
Sub-family (consumption axis):
if C ≥ 0 then G ← Consumptive
else if C < 0 and φ > 90 then G ← Ascetic
elseG ← Austerity
end if
return (P, W, C, Octant, Δ, Σ, Γ, Ξ, ρ, θ, φ, F, G)
```

The time complexity remains $\mathcal{O}(k)$ per organization and $\mathcal{O}(nk)$ for a batch of n organizations. The sub-family decision tree on lines 20–27 adds only $\mathcal{O}(1)$ operations.

6 Machine Learning and Data Science

6.1 Multi-Output Regression

We now estimate three outputs (P, W, C) simultaneously. Let the feature vector $\mathbf{x} \in \mathbb{R}^d$ contain observable proxy variables. Define the regression:

$$\begin{pmatrix} P \\ W \\ C \end{pmatrix} = \mathbf{B}^\top \mathbf{x} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}_3(\mathbf{0}, \boldsymbol{\Omega}), \tag{20}$$

where $\mathbf{B} \in \mathbb{R}^{d\times 3}$ is the coefficient matrix and $\boldsymbol{\Omega} \in \mathbb{R}^{3\times 3}$ is the residual covariance. The joint covariance structure captures the empirical fact that productivity, wages, and consumption are not independent: in most economies $\mathrm{Cov}(P, W) > 0$ and $\mathrm{Cov}(W, C) > 0$ (higher wages drive higher consumption).

The GLS estimator is:

$$\hat{\mathbf{B}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{Y}, \tag{21}$$

where $\mathbf{Y} \in \mathbb{R}^{n\times 3}$ is the matrix of observed (P_i, W_i, C_i) values and $\mathbf{X} \in \mathbb{R}^{n\times d}$ is the design matrix.

6.2 Multiclass Octant Classification

With eight octants, the classification task becomes an 8-class problem. The natural decision boundaries are the three planes $P = 0$, $W = 0$, $C = 0$, all of which are *axis-aligned hyperplanes* in the feature space (after regression to the (P, W, C) subspace). Hence:

(i) An *oblique random forest* trained on $(\hat{P}, \hat{W}, \hat{C})$ achieves Bayes-optimal classification, since the true boundaries are hyperplanes.

(ii) A three-stage *hierarchical binary classifier* first separates $P > 0$ vs. $P < 0$, then $W > 0$ vs. $W < 0$ within each branch, then $C > 0$ vs. $C < 0$, for a total of three binary decisions with interpretable thresholds.

6.3 Clustering in $\mathcal{K}$

k-means with $k = 8$ clusters and initializations at the eight corners $(\pm 1/2, \pm 1/2, \pm 1/2)$ minimizes:

$$\mathcal{J} = \sum_{o=1}^{8} \sum_{\mathbf{v}\in C_o} \|\mathbf{v} - \boldsymbol{\mu}_o\|_2^2, \tag{22}$$

where $\boldsymbol{\mu}_o \in \mathbb{R}^3$. Convergence is rapid because the eight natural clusters are convex, mutually disjoint, and separated by coordinate hyperplanes.

6.4 Neural Embedding to $\mathcal{K}$

Extending the two-dimensional embedding, we train a transformer encoder $f_\phi : \mathcal{X} \to \mathbb{R}^d$ with a three-output regression head $h_\psi : \mathbb{R}^d \to [-1, 1]^3$ using tanh activations:

$$\mathcal{L}(\phi, \psi) = \frac{1}{n} \sum_{i=1}^{n} \left\| h_\psi\Big(f_\phi(\mathbf{x}^{(i)})\Big) - (P^{(i)}, W^{(i)}, C^{(i)})^\top \right\|_2^2. \tag{23}$$

The three output dimensions now have distinct economic interpretations enabling targeted interpretability analysis: attention attribution can separately identify productivity-driving, wage-determining, and consumption-influencing textual features in organizational descriptions.

Disentanglement objective. Because $\mathrm{Cov}(P, W, C) \neq \mathbf{0}$ empirically, a *disentangled* encoder is desirable: we add an independence regularizer to (23),

$$\mathcal{L}_{\mathrm{reg}}(\phi, \psi) = \mathcal{L}(\phi, \psi) + \lambda \left\| \hat{\mathbf{\Omega}} \right\|_F^2, \tag{24}$$

where $\hat{\mathbf{\Omega}}$ is the empirical covariance of $(h_\psi(f_\phi(\mathbf{x}^{(i)})))_{i=1}^n$ and $\lambda > 0$ penalizes correlation between the three output dimensions, encouraging the model to learn *independent* organizational axes.

7 Visualizations

7.1 Three-Dimensional P–W–C Cube

Figure 1 illustrates the unit cube $\mathcal{K}$ with the eight octants colour-coded, the fair-exchange ray, and selected organizational positions.

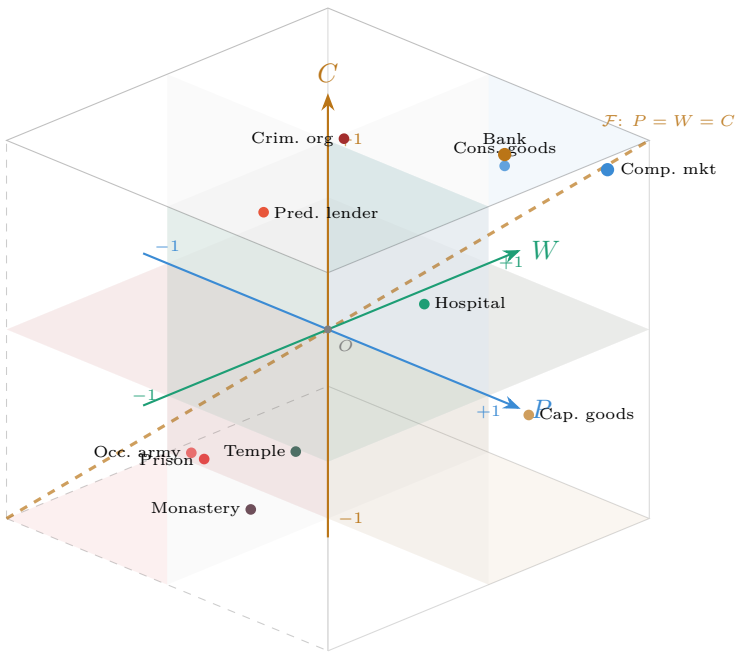
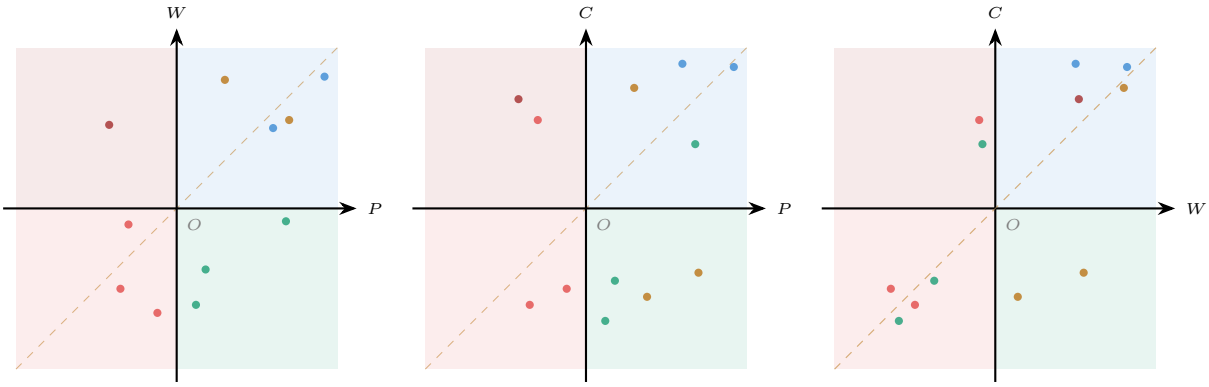


Figure 1: The P–W–C unit cube $\mathcal{K}$ with selected organizational positions.

The dashed amber line is the fair-exchange ray $\mathcal{F} : P = W = C$. Octants are lightly shaded: blue ($\mathcal{O}_{+++}$), teal ($\mathcal{O}_{+-+}$), amber ($\mathcal{O}_{++-}$), red ($\mathcal{O}_{---}$).

7.2 Projected Sections

Figure 2 shows three two-dimensional projections of the P–W–C cube, one for each coordinate pair, allowing organizations to be compared along any two axes independently.



(a) P–W projection (b) P–C projection (c) W–C projection (dashed = $\Gamma = 0$)

Figure 2: Three two-dimensional projections of the P–W–C organizations.

The amber dashed diagonal in each panel is the projection of the fair-exchange ray: $W = P$ (left), $C = P$ (centre), $C = W$ i.e. $\Gamma = 0$ (right). Colors: blue ($\mathcal{O}_{+++/++-}$), teal ($\mathcal{O}_{+-\pm}$), amber (banks and capital-goods), red (parasitic).

7.3 Consumption Elevation φ by Archetype

Figure 4 plots the consumption elevation angle φ on the horizontal axis against θ on the vertical axis, producing a two-angle archetype map. Each organization is represented by a point (θ, φ), and the three super-families occupy distinct regions of this map.

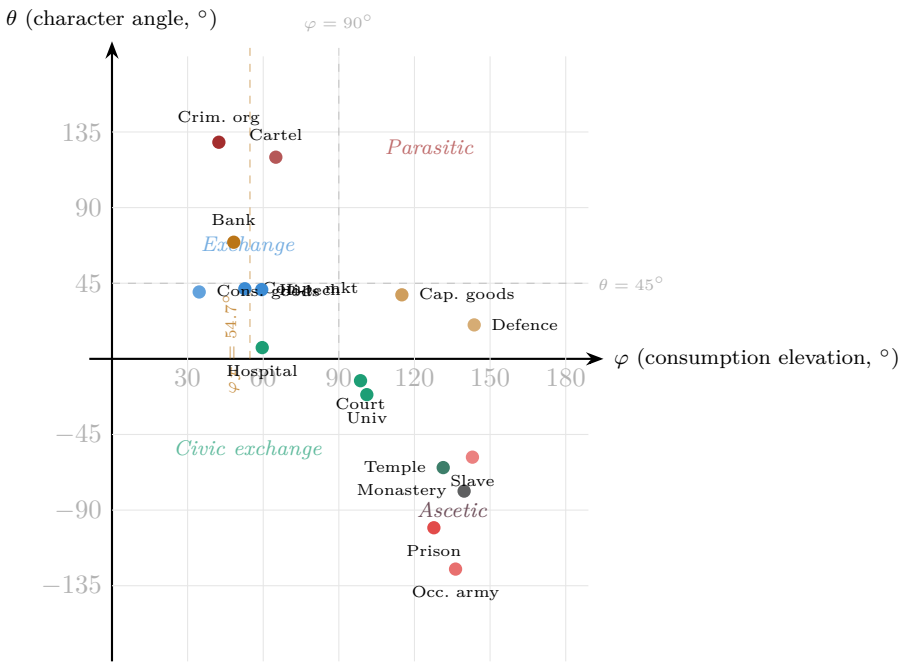


Figure 4: Two-angle archetype map (φ, θ).

Each point is an organization; the vertical dashed line at $\varphi = 90°$ separates consumptive ($\varphi < 90°$) from ascetic/austerity ($\varphi > 90°$) organizations. The horizontal dashed line at $\theta = 45°$ is the fair-exchange direction. The amber vertical at $\varphi \approx 54.7°$ is the projection of the fair-exchange ray $\mathcal{F}$.

8 Economic Interpretation

8.1 The Consumption Axis and Aggregate Demand

In Keynesian macroeconomics, aggregate demand is AD $= C + I + G + (X - M)$. The P–W–C framework maps the first component directly: the consumption axis C of an organization captures its net contribution to final demand. Institutions with $C > 0$ (consumer-goods firms, banks through credit, welfare states) *sustain* the Keynesian multiplier; institutions with $C < 0$ (monasteries, capital-goods firms, prisons) *contract* it.

Proposition 8.1 (National Aggregate Demand Identity)**.** *Let $\{(P_i, W_i, C_i)\}_{i=1}^m$ be the organizational portfolio of a nation with employment shares $\{s_i\}$. The weighted mean consumption is:*

$$\bar{C} = \sum_{i=1}^{m} s_i C_i, \tag{25}$$

and the nation's position on the Keynesian demand spectrum is determined by the sign of $\bar{C}$. A nation dominated by ascetic or capital-goods organizations ($\bar{C} < 0$) faces secular demand deficiency; one dominated by consumer-goods and finance ($\bar{C} > 0$) faces potential demand inflation.

8.2 The Consumption–Wage Gap and Inequality

The consumption gap $\Gamma = C - W$ captures the *Veblen effect* at organizational level [5]: whether organizational membership drives consumption *beyond* what wages alone would support. When $\Gamma > 0$ (as in banks with $\Gamma = -0.05$ or consumer-goods firms with $\Gamma = 0.40$), credit and status competition generate consumption in excess of earned income. When $\Gamma < 0$ (monasteries with $\Gamma = -0.10$, capital-goods firms), participants consume below their wage, generating investable surplus.

8.3 Three-Dimensional Welfare and the Triad

The national welfare identity now extends to:

$$\mathcal{W}_{\text{nat}} = \sum_{i=1}^{m} s_i \, \Xi_i = \sum_{i=1}^{m} s_i (P_i + W_i + C_i), \tag{26}$$

where $\Xi_i = P_i + W_i + C_i$ is the triad welfare of organization i. This extends the two-dimensional net social value Σ by incorporating the demand-side contribution. A temple with $\Sigma = -0.20$ (below zero in two dimensions) may nonetheless contribute positively to national welfare if its C value captures social cohesion effects on consumption.

9 Discussion and Limitations

The addition of the C axis greatly enriches the framework but introduces new measurement challenges.

Defining consumption attribution. Attribution of consumption to a specific organization is theoretically tractable (a consumer-goods firm's output becomes final demand) but empirically complex for financial intermediaries and public institutions. We advocate the *induced consumption* concept: the total additional final demand generated by one additional unit of the organization's activity, estimated via input–output tables.

Negative consumption. $C < 0$ requires careful interpretation. It does not mean an organization physically removes goods from the economy, but that it *suppresses* final demand relative to the counterfactual. A monastery suppresses the consumption that its members would otherwise undertake; a prison prevents incarcerated individuals from participating in consumer markets.

Independence of axes. The three axes P, W, C are empirically correlated across organizations. In most economies, $\mathrm{Cov}(W, C) > 0$ (higher wages generate higher consumption) and $\mathrm{Cov}(P, W) > 0$ (more productive firms pay more). The framework does not assume independence; the ML disentanglement objective (24) explicitly accounts for residual correlation.

10 Conclusion

The extension of the P–W framework to the three-dimensional P–W–C cube $\mathcal{K}$ yields a richer and more complete account of organizational structure. Key results:

- The unit cube has **eight octants** encoding all combinations of value creation/destruction, labor distribution, and demand stimulation/suppression.
- The **fair-exchange ray** $P = W = C$ generalizes the two-dimensional fair-exchange locus, with the precise angular coordinates $\theta = 45°$ and $\varphi \approx 54.74°$.
- The two angles (θ, φ) together form a **sufficient statistic** for organizational archetype, independent of scale.
- The new metric $\Gamma = C - W$ captures the **Veblen–Keynes gap**: whether consumption exceeds wages (credit economy) or falls short (austerity/asceticism).
- The **triad welfare** $\Xi = P + W + C$ extends social welfare accounting to include demand-side effects.
- The framework connects naturally to **Keynesian aggregate demand**, **input–output analysis**, and **transformer-based neural embedding** for large-scale automated institutional classification.

References

[1] S. Ghosh, "The Unified P-W Framework for the Taxonomy, Analysis, and Classification of Organizations," Kolkata, 2026.

[2] A. Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations.* W. Strahan and T. Cadell, London, 1776.

[3] J. M. Keynes, *The General Theory of Employment, Interest and Money.* Macmillan, London, 1936.

[4] K. Marx, *Das Kapital: Kritik der politischen Ökonomie*, vol. 1. Otto Meissner, Hamburg, 1867.

[5] T. Veblen, *The Theory of the Leisure Class: An Economic Study of Institutions.* Macmillan, New York, 1899.

[6] P. A. Samuelson, "The pure theory of public expenditure," *Review of Economics and Statistics*, vol. 36, no. 4, pp. 387–389, 1954.

[7] O. E. Williamson, *The Economic Institutions of Capitalism.* Free Press, New York, 1985.

[8] T. Piketty, *Capital in the Twenty-First Century.* Harvard University Press, Cambridge, MA, 2014.

[9] D. Acemoglu and J. A. Robinson, *Why Nations Fail.* Crown Publishers, New York, 2012.

[10] W. Godley and M. Lavoie, *Monetary Economics.* Palgrave Macmillan, Basingstoke, 2007.

[11] H. P. Minsky, *Stabilizing an Unstable Economy.* Yale University Press, New Haven, 1986.

[12] D. C. North, *Institutions, Institutional Change and Economic Performance.* Cambridge University Press, Cambridge, 1990.

[13] W. W. Leontief, *The Structure of the American Economy, 1919–1929.* Harvard University Press, Cambridge, MA, 1941.

Glossary

P **(normalized productivity)**
Ratio Net Productivity/$P_{\max} \in [-1, 1]$. Positive: net value creation. Negative: net value destruction.

W **(normalized wage)**
Ratio Mean Compensation/$W_{\max} \in [-1, 1]$. Positive: net income to participants. Negative: net transfers from participants (fees, tithes, forced labor).

C **(normalized consumption)**
Ratio Net Consumption Generated/$C_{\max} \in [-1, 1]$. Positive: net stimulation of final demand (consumer goods, credit). Negative: net suppression of demand (prisons, monasteries, wartime rationing).

Δ **(surplus extraction)**
Scalar $\Delta = P - W$. Positive: labor surplus extracted by the organization. Negative: participants paid above marginal product (rent-sustained).

Σ **(productivity–wage net value)**
Scalar $\Sigma = P + W$. Total welfare from production and labor distribution. As in the two-dimensional framework.

Γ **(consumption gap)**
Scalar $\Gamma = C - W$. Positive: participants consume more than they earn (credit, status, transfers). Negative: participants consume less than they earn (forced saving, asceticism). Captures the Veblen–Keynes demand gap at organizational level.

Ξ **(triad welfare)**
Scalar $\Xi = P + W + C$. Total welfare impact including demand-side effects. The three-dimensional extension of Σ.

ρ **(organizational magnitude)**
Euclidean norm $\rho = \sqrt{P^2 + W^2 + C^2} \in [0, \sqrt{3}]$. Scale of the organization's total influence. $\rho = \sqrt{3}$ at the cube corners.

θ (character angle)
Azimuthal angle $\theta = \arctan(W/P) \in (-180°, 180°]$. Classifies productivity–wage archetype. Unchanged from the two-dimensional framework.

φ (consumption elevation)
Polar angle $\varphi = \arccos(C/\rho) \in [0°, 180°]$. $\varphi < 90°$: consumptive organization. $\varphi = 90°$: consumption-neutral. $\varphi > 90°$: ascetic or demand-suppressing.

$\mathcal{K}$ (P–W–C space)
Unit cube $[-1, 1]^3 \subset \mathbb{R}^3$ with axes P, W, C. The three-dimensional extension of the P–W plane.

Fair-exchange ray ($\mathcal{F}$)
Main diagonal $\{(t, t, t) : t \in [-1, 1]\}$ of $\mathcal{K}$. Characterized by $\Delta = 0$, $\Gamma = 0$, $\Xi = 3P$, $\theta = 45°$, $\varphi \approx 54.74°$. Generalizes the two-dimensional fair-exchange locus $W = P$.

Octant ($\mathcal{O}_{s_1 s_2 s_3}$) One of the eight open regions of $\mathcal{K}$ defined by the signs $(s_1, s_2, s_3) = (\operatorname{sgn} P, \operatorname{sgn} W, \operatorname{sgn} C) \in \{+, -\}^3$. Generalizes the four quadrants of the two-dimensional framework.

Triad welfare (Ξ)
See Ξ above. Weights the demand-side contribution C equally with production P and distribution W.

Consumption gap (Γ)
See Γ above.

Super-family One of three broad institutional classes determined by the signs of P and W: Exchange ($P > 0$, $W > 0$), Civic ($P > 0$, $W < 0$), Parasitic ($P < 0$).

Sub-family (consumption)
Within each super-family: Consumptive ($C > 0$), Austerity ($C < 0$, $\varphi \leq 90°$), or Ascetic ($C < 0$, $\varphi > 90°$, indicating elevated consumption suppression).

The End